RECOMMENDATION SYSTEMS

PACF

- Agenda ⇒
  - Market Basket Analysis
  - Apriori Algo, Association Rule mining
    (Lift, Support, confidence)

IDS

Walmart | DMART

↑ sales & Customer satisfaction

identify items purchased together

items ⇒ Placed under same shelf

kind

- improve Buying capacity

X → Y

PS → Supermarket store

(Grocery shopping)

→ optimise product placements

→ items → purchased together

→ Bundle items → combo

Bread + Butter

Bread + Milk

Solve → Market Basket Analysis

↓

Apriori

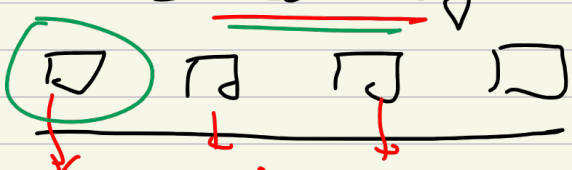
Amazon → frequently Bought together

item → similars

iphone → charger

(2) People who Bought → [ ]

also Bought



Charger, apple cable, Headphone

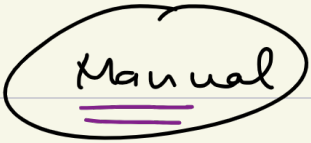
(4) list of 4 to 5 items

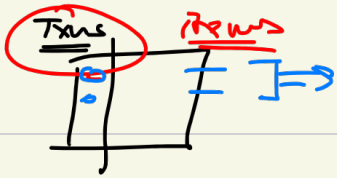
insurance
screen protector

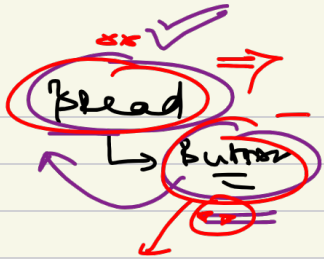
idea → old

legacy → APRIORI

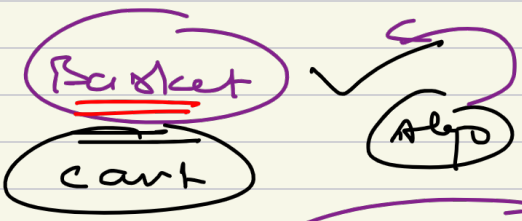
So →



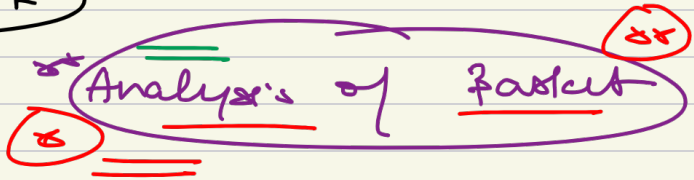


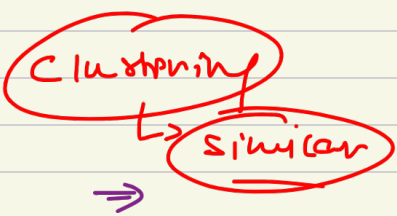


(x) Scale of data ↑



Grocery





$\Rightarrow$

$items = \{ Milk, Bread, Jam, Butter -- \}$

$D = \{ 1, 2, 3 ---- n \}$ items

$T = \{ T_1 \quad T_2 --- T_m \}$ Txns

$m \gg n$

500, 1000, 10000

m → Txns

daily → 100 Txns

2 years → 700 × 100

Txns — Bought Together

$T_1 = \{1, 3, 6, 8\}$

$T_2 = \{1, 3, 7, 12\}$

$T_3 = \{1, 7, 3\}$

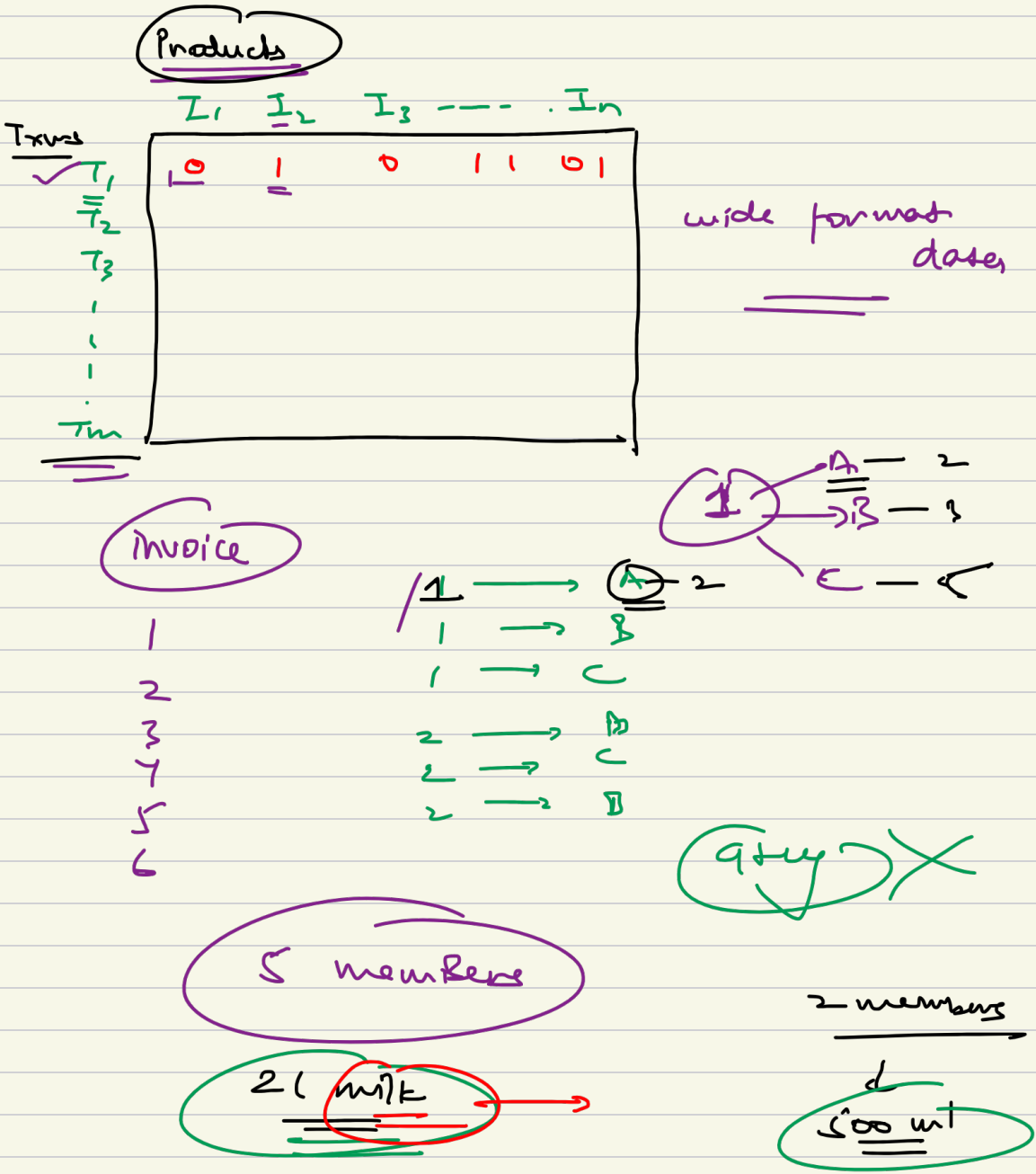
$\{1, 3\}$ → milk, Bread

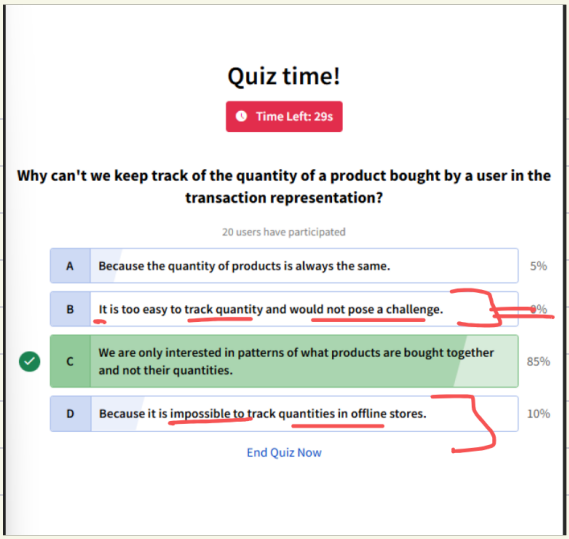
$\{1, 3\} \rightarrow 3$

$\{1\} \longrightarrow \{3\}$ 3

frequency of $\{1, 3\}$

P(milk & Bread) >> P(milk & Toy car)


Products
I1 I2 I3 --- In
Txns
T1
T2
T3
Tm
wide format data
Invoice
1
2
3
4
5
6
1 → A 2
1 → B
1 → C
2 → D
2 → C
2 → D
5 members
2 members
2 milk
500 ml



⇒

Data Preprocessing ⇒ ✓

Apriori → frequent patterns

(items bought together)

$D = \{1, 3, 6, 7, 8, 12, 16\}$

⇒ items

Txns

$T_1 = \{1, 3, 6, 8\}$

$T_2 = \{1, 3, 7, 12\}$

$T_3 = \{1, 7, 3, 16\}$

DATA STRUCTURE

(key, value)

item freq

key	value
{1}	3
{3}	3
{6}	1
{8}	1

{1,3} : 3
{1,7} : 2
{1,3,7} : 2

frequency map

* Possible subsets ⇒

100 items

n = 100

Products

Total possible subsets = $2^n - 1$

{}

4 items

2^4

$\Rightarrow 16 - 1 = 15$

2

$2^2 - 1$

* (A, B) — {A}, {B}, {A, B}, { } ✗

$2^2 - 1 = 3$

$2^{\uparrow} - 1$

10000 ⇒ subsets?

Big ↑

Method → NOT scalable

$2^{10000} - 1$:

set → min-support → frequency

→ Threshold

min_threshold = 2

c = 2

freq.

superset ≥ min_threshold

Support {1} : 3

{3} : 3

{7} : 2

{6} : 1 {8} : 1 {12} : 1

support

threshold

&

min_support

Support = 2

{6} : 1

{6, 2}

{6, 7}

{6, 12}

{7}

{7}

n = 7

Total subsets = $2^7 - 1$ = 127

{9}

{10, 12}

only take combⁿ which are above threshold

min_support = ✓

(key, value)

item freq

3 mins

10:10

100 Txn

Bread → 60 Txn

Support = 60/100 = 0.6

Milk = 40/100 = 0.4

30 → 0.3

min — support = 0.4

A = {2, 4}

< 100

1

Quiz time!

Time Left: 13s

Consider an item-set: $A = \{2, 4\}$. Let the number of times this item set occurs in the transaction data be < 100

Then, how many times would ~~a subset~~ of A, let $B = \{2, 4, 6\}$ occur in T?

17 users have participated

A less than 100 76%

B greater than 100 18%

C equal to 100 0%

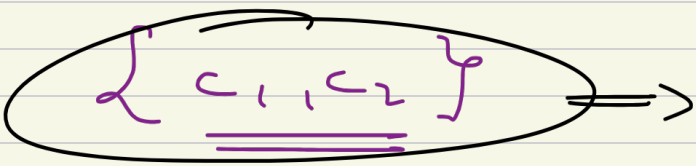
D Can't say 6%

End Quiz Now

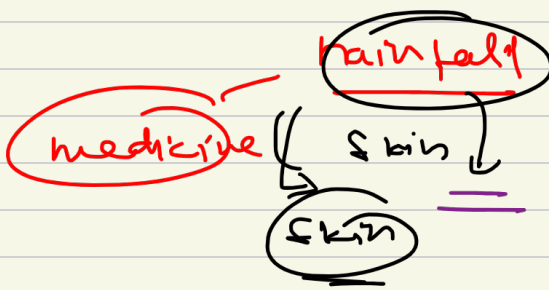
Apriori

(1) Bioinformatics

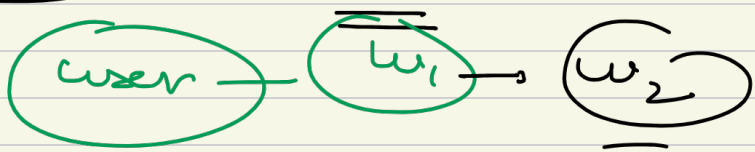
chemical compounds



(2) medicines



(3) web page → w_1 → w_2



Apriori → frequency

→ prob.

Support

* multiple itemsets

support

Association Rule mining

↳ come up with rules

X → what to suggest

Buy

Apriori → support

X = { 1, 2, 3 } Y = { 4, 6 }

consequents

antecedents

{ 1, 2, 3 } → { 4, 6 }

{ milk, eggs } → { Bread, Butter }

{7, 9}

Apriori

{1, 2, 3, 4, 6}

high freq

Support

Evaluate

Association metrics

1. Support (x) — milk

= # of subset milk (x) occured / Total Trns

= % age

P(x in T)

2. Confidence (X → Y)

milk

bread

$= \frac{\text{Txns} \rightarrow \text{Both X \& Y appear}}{\text{how many Txn with X}}$

$P(\text{Bread}/\text{Milk}) = \frac{P(B \cap M)}{P(\text{Milk})}$

Confidence $\Rightarrow$ $X \rightarrow Y$

(3) Lift

grocery shop

four Txns

Dairy $\rightarrow$

{cow, lakes} $\rightarrow$ milk

{yogurt} $\rightarrow$ {milk}

{T B} $\rightarrow$ {milk}

Total Txn = 100

Milk → 80 Txn

{milk}

TB → 14 Txn

TB + Milk → 8 Txn

{milk, Toothbrush}

80%

$$P(Milk \mid TB) = \frac{P(TB \cap M)}{P(TB)}$$

$$= \frac{8}{14} = 57\%$$

TB

57% < 80%

80%

Lift

Lift (TB → Milk)

Confidence

$$= \frac{P(TB \cap Milk)}{P(TB)\ P(Milk)}$$

$$= \frac{8}{14.80} = 0.71$$

Lift > 1 ⇒ TB Buyers more likely to Buy milk

Lift = 1 ⇒ TB & Milk

Lift < 1 ⇒ TB less likely to Milk

Conf.

TB → milk

Lift

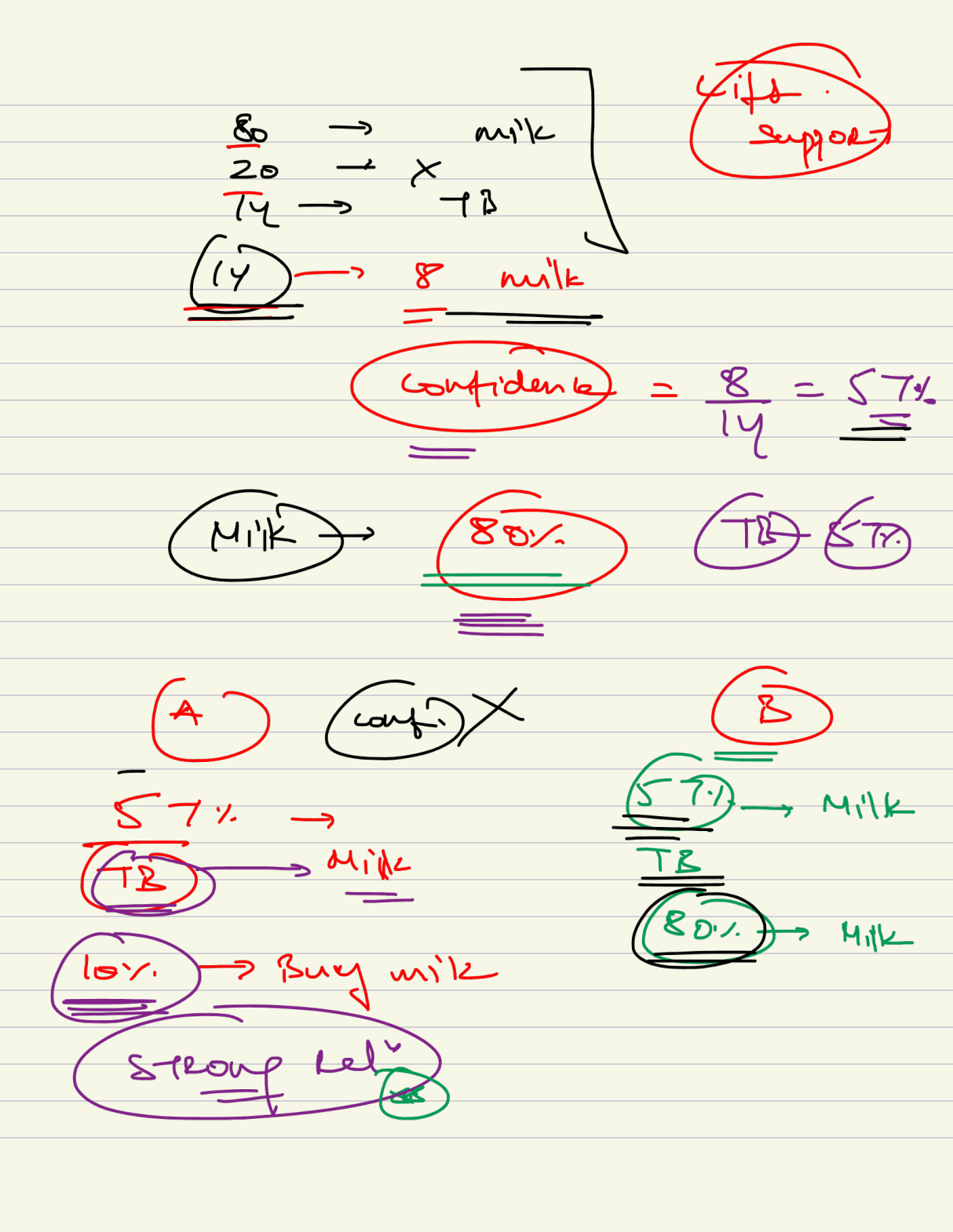
TB

Milk

10-15 mins

Implementation

Rules ⇒


80 → milk
20 → X
TY → TB
(1Y) → 8 milk
Confidence = 8/1Y = 57%
Milk → 80%
TB → 57%
A
conf. X
57% →
TB → Milk
10% → Buy milk
Strong rel
B
57% → Milk
TB
80% → Milk

Lift → TB

Normally happens

Lift (TB → Milk)

Conf: → Y. egg

57% → TB → Milk

impression

STORE A

100

10 → Milk

TB → 8/14

Buy milk

TB → Milk = 57%

STORE B

100

90 → Milk

TB 8/14

TB → 57%

Confidence

Does TB actually make milk more likely

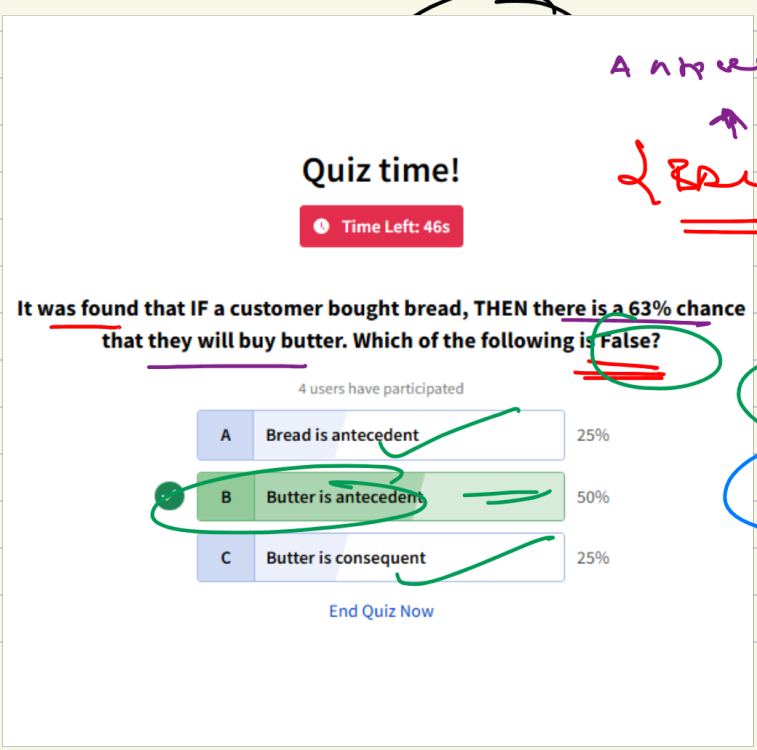
Left

Milk

80%

TB

Confidence = 5 Tx. / P(Milk) = 0.8



Antecedent

{Bread} → {Butter}

20 mins

Invited

Business